Module 7 Discussion: Future of AI

The area I keep coming back to is AI in finance, mainly portfolio balancing and fraud detection. I'm working toward a career in the retirement industry, so this one hits close to home. A regular computer uses bits, which are either a 0 or a 1. A quantum computer uses qubits, and a qubit can be in superposition, which basically means it can be a mix of 0 and 1 at the same time. Two qubits can hold all four 2-bit combinations, three can hold eight, and it keeps growing from there (IBM, 2025). Qubits can also be entangled, so measuring one tells you something about the others. Balancing a portfolio means weighing a ton of variables at once, which is the kind of messy problem quantum is supposed to be good at. IBM lists financial market modeling and pattern finding as likely uses and says quantum might speed up some machine learning problems. That could mean AI models that train and run faster.

Honestly, I'm both excited and a little worried. In Module 6 I wrote about retirement fraud, so AI catching suspicious activity on retirement accounts faster really excites me. My bigger concern is security. NIST says a powerful enough quantum computer could break the encryption that protects just about everything we do online, and some experts think that could happen within a decade (NIST, 2024). That includes bank accounts and 401(k)s. There's also a risk called "harvest now, decrypt later," where criminals steal encrypted data today and wait until they can unlock it. CISA calls this out for the financial sector (CISA, n.d.). The good news is NIST released its first three post-quantum encryption standards in August 2024.

There are still big challenges. Qubits are fragile. Heat and noise cause errors, so the processors have to be kept colder than outer space, close to absolute zero (IBM, 2025). Error correction is another hurdle since you need extra qubits just to fix mistakes. The full system is about the size of a car, so cost is a real issue. I also don't think many people are trained in this yet. And companies have to switch to quantum-safe encryption, which NIST says will take time.

So will quantum computing have a major impact in the next 20 years? I think yes, just not the way people picture it. I don't see regular people buying a quantum computer. IBM says they are anything but portable. The impact will come through big companies, government, and security, plus medicine and materials, where quantum could speed up drug research (IBM, 2025). Most of us will feel it indirectly, like better fraud alerts, new medicines, or our bank quietly upgrading its encryption. Since I'm heading into the retirement industry, the security side is what I'll be watching.

References:

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